

INES AITSAHALIA

PHD CANDIDATE

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NEW YORK CITY, NY

EDUCATION

2022-Present (est. 2027) **Columbia University**, Center for Theoretical Neuroscience

PhD in Neurobiology and Behavior

MPhil in Neurobiology and Behavior (2025)

MA in Neurobiology and Behavior (2024)

2018-2022 **Princeton University**

BA in Neuroscience, Cognitive Science

Graduated with Honors and John Brinster, Class of 1943, Senior Thesis Prize

RESEARCH EXPERIENCE

2024-Present **Iigaya Lab & Stachenfeld Lab**, Columbia University

PhD Candidate

2025 **BASIS AI Research Institute**,

PhD Research Intern

2023 **Miller Lab**, Center for Theoretical Neuroscience at Columbia University

PhD Student Researcher (Rotation)

2022 **Horga Lab**, Columbia University and New York State Psychiatric Institute

PhD Student Researcher(Rotation)

2022 **Iigaya Lab**, Columbia University and New York State Psychiatric Institute

PhD Student Researcher (Rotation)

2020-2022 **Niv Lab**, Princeton University

Undergraduate Research Assistant

Summer Research Intern

PUBLICATIONS & PRESENTATIONS

A flexible hierarchical clustering model for cognitive abstraction and generalization in psychiatry

I. Aitsahalia & K. Iigaya

2025, in prep

Sparse Cognitive Graphs: Linking dopamine to cognitive map through reinforcement learning

A. Gungi, P. Sepulveda, I. Aitsahalia, K. Iigaya

2025, in prep

“Learning from Outliers” I. Aitsahalia	2025 Talk at NY Memory Hub
Inferring cognitive strategies from groups of animals in natural environments I. Aitsahalia, T. Botch, S. Gu, T. O’Connell, R. Seigel, R. Peterson, D. Batenkov, E. Mackevicius	2025 NeurIPS workshop paper
Anticipatory value computation as a new transdiagnostic target P. Sepulveda, I. Aitsahalia, K. Kumar, T. Atkin, K. Iigaya	2024, Biological Psychiatry
“A multi-level latent cause framework for the study of delusions” I. Aitsahalia, G. Horga, K. Iigaya	2025, Poster at Computational Psychiatry Conference (CPC)
Building Deterministic Causal Graphs Using Reinforcement Learning in Cognitive Tasks with Evolutionary Bias I. Aitsahalia, A. Gungi, P. Sepulveda, K. Iigaya	2024, Computational Cognitive Neuroscience
Book Chapter: Computational Psychiatry and Eating Disorders I. Aitsahalia*, P. Sepulveda*, S. Dong*, K. Iigaya (*denotes co-first authors)	2024, Oxford Handbook on Eating Disorders
“Unified anticipation model can explain information-seeking around negative outcomes” I. Aitsahalia, P. Sepulveda, V. Fisher, S. Dong, K. Iigaya	2024, Poster at CPC
“Common cause inference as a generative model of delusions” M. Schafer, A. Velikovsky, I. Aitsahalia, K. Iigaya, G. Horga 2024 poster	2024, Poster at CPC
“Bridging reinforcement learning and causal inference: Emergence of grid cells in abstract tasks” A. Gungi, P. Sepulveda, I. Aitsahalia, M. Schaffer, K. Iigaya	2024, Poster at CoSyNe
“Information seeking around aversive outcome anticipation” I. Aitsahalia, P. Sepulveda, S. Dong, R. Chen, Y. Chai, K. Iigaya	2023, Poster at SfN
“Computation of Anticipatory Value as a New Transdiagnostic Target” I. Aitsahalia, P. Sepulveda, K. Kumar, K. Iigaya	2023, Poster at CPC
“Alterations in Perceptual Learning Drive Speech Hallucinations in Medication-Free Patients with Schizophrenia” J. Buck, I. Aitsahalia, D. Serrano, G. Horga	2023, Poster at CPC
“Why do some beliefs and behaviors resist updating?” S. Pisupati, I. Aitsahalia, S.N. Quinard, I.M. Berwian, A.J. Langdon, Y. Niv	2023, Poster at CPC
“Latent-cause inference under varying beliefs about randomness captures individual differences in fear extinction” I. Aitsahalia, S. Pisupati, S.N. Quinard, I.M. Berwian, Y. Niv	2022, Poster at SfN
“Over- and under-segmentation of latent causes lead to different types of extinction failures” I. Aitsahalia, S. Pisupati, S.N. Quinard, I.M. Berwian, Y. Niv	2021, Poster at SfN

TEACHING & SERVICE

2025 **Committee on Rigor, Neurobiology and Behavior** Columbia University

2024 **Preventing and Managing PTSD, certification**

2023, 2024, 2025 **Course Instructor, Math Tools for Theoretical Neuroscience**

Columbia University

2023-2024 **Course Instructor, Mind/Brain Disorders** Barnard College

2023, 2025 **Teaching Assistant, Analysis for Neuroscientists** Columbia University

2023-2025 **Teaching Assistant, Introduction to Cognitive Science** Barnard College

2022-Present **Women on Two Floors**

Affinity group for women in computational neuroscience at Columbia

2021-2022 **Subcommittee on Inclusive Teaching** Princeton University

2022-Present **Dragonfly Mental Health**

2021-2022 **Empowering Diversity and Promoting Scientific Equity at PNI** Princeton University

AWARDS

2025 **Best Data Blitz** New York Memory Hub Conference

2025 **Student Leadership and Service Award** Columbia University

2025 **Presidential Teaching Award Finalist** Columbia University

2024-2025 **Lead Teaching Fellow**, Columbia University Center for Teaching and Learning

2022-Present **NSF Graduate Research Fellow**, Cognitive Science

2022 **Knox Studentship**, Cambridge University *Offer declined*

2022 **Rhodes Scholarship semifinalist**, Oxford University

2022 **John Brinster, Class of 1943, Senior Thesis Prize**, Princeton University

2022 **Lewis Center Class Day Award**, Princeton University

2021 **Nancy J. Newman, MD '78 & Valerie Biousse, MD Research Fund for Neuroscience**,
Princeton University

2021 **Program in Cognitive Science Summer Research Award**, Princeton University